Top Natural-Energy Home Products of 2025

Corrected edition with product details and prices · Updated October 2026

Based on 2025 market trends, sustainability reports and consumer reviews, the leading natural and renewable energy products focus on home electrification, AI-assisted efficiency, and using solar or geothermal energy to reduce grid dependence.

Here are 16 popular, well-reviewed products for using natural energy at home. Each entry lists the manufacturer, main components, operating principle, typical U.S. price and what installation requires, with a diagram of how it works. Prices are typical 2026 figures and vary by retailer, region and installation. Installation notes are general guidance: always follow the manufacturer's instructions and local codes.

1. Solar-Assisted Heat Pump Water Heater

These units combine an outdoor solar-thermal collector panel with heat pump technology, drawing heat from both sunlight and the surrounding air to produce hot water at a lower running cost than a standard electric tank. Smart Solar is one leading example; Eco-$mart also sells a U.S.-made solar-assisted model.

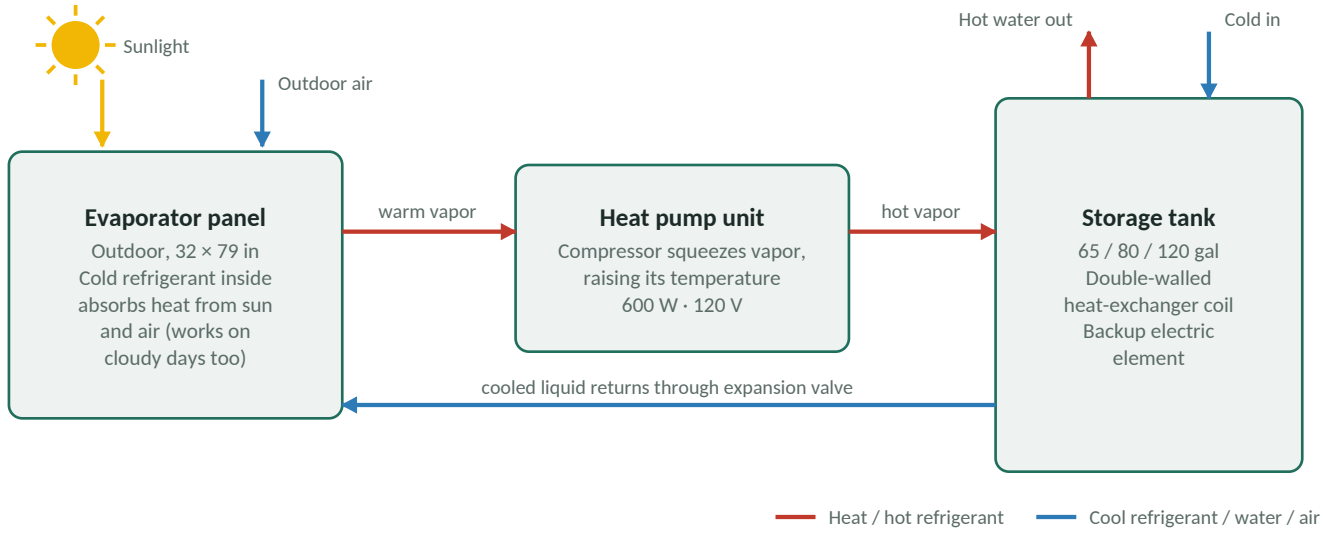


Solar-assisted heat pump water heater: storage tank with evaporator panel.

Manufacturer	Smart Solar (sold and installed in New England, e.g. by New England Solar Hot Water). Alternative: Eco-$mart (ecosmartinc.com), “Solar Assisted Heat Pump Water Heater (U.S. Made).”
Components	Storage tank (65, 80 or 120 gal); compact heat pump unit (12 × 12 × 24 in); outdoor evaporator panel (32 × 79 in) that the refrigerant runs through; double-walled potable heat exchanger; optional 120 V or 240 V backup electric element.
How it works	Refrigerant is pumped through the outdoor panel, where it absorbs heat from sunlight and ambient air. The heat pump compresses the refrigerant to raise its temperature and transfers that heat into the tank through the heat exchanger. Because it collects heat rather than generating it, it uses far less electricity than a resistance heater.

Key specs	Annualized COP 2.7 (Boston); UEF 2.8 (ENERGY STAR); heat pump draws 600 W on a 120 V/15 A circuit; ~8,000 BTU/h capacity; operates from −10 °F to 110 °F outdoors; 48 dBA.
Typical price	About **$6,700 installed** for an 80-gallon system before incentives (Massachusetts Clean Energy Center example: $6,725; $5,225 after the Mass Save rebate).
Installation needs	Professional install by a plumbing/HVAC contractor experienced with split heat pump systems. Needs: a sunny outdoor wall or roof spot for the 32 × 79 in panel; refrigerant lines between panel, heat pump and tank; a 120 V/15 A circuit (a 240 V circuit only if you choose the larger backup element); and room for the tank (it fits closets or small utility rooms, so no basement is needed). Expect plumbing and electrical permits.

Figure 1. Components and how Solar-Assisted Heat Pump Water Heater works



2. Tesla Powerwall 3 (and similar home batteries)

A home battery with a built-in solar inverter. It stores solar energy and can shift household usage away from peak-price grid hours and provide backup during outages. How much it reduces grid use depends on the size of the solar system, household consumption and climate.

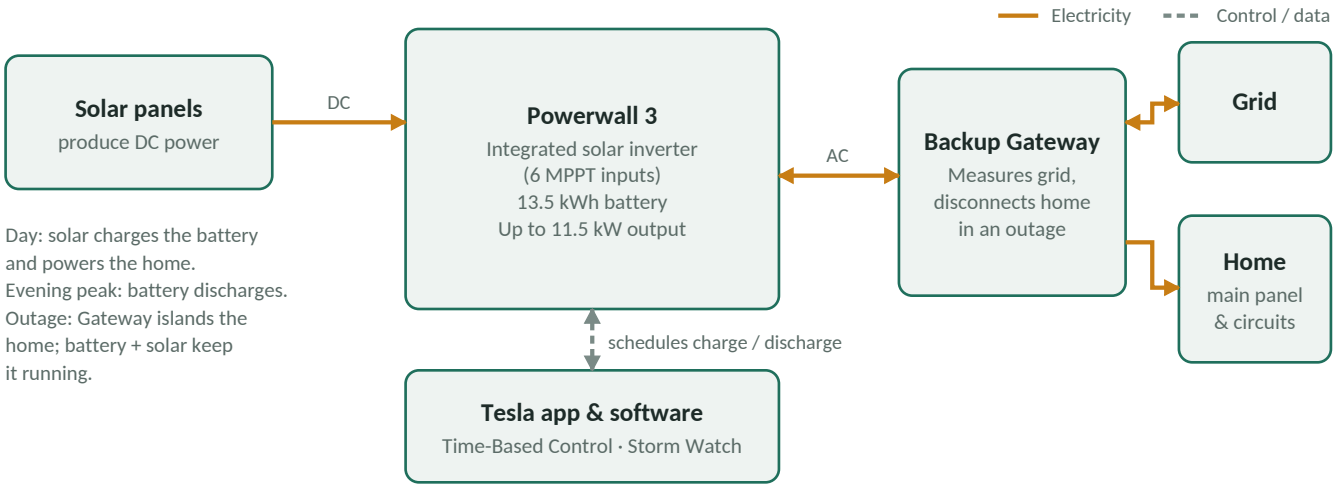


Powerwall units installed on a garage wall, next to an EV charger.

Manufacturer	Tesla, Inc.
Components	13.5 kWh lithium-ion battery pack; integrated hybrid solar inverter with 6 MPPT inputs (up to 20 kW DC solar); Tesla Backup Gateway (required for backup; switches the home off the grid during an outage); Tesla app; optional expansion packs.
How it works	DC-coupled: solar power charges the battery directly or is inverted to AC for the home. Software (time-based control, Storm Watch) decides when to charge, when to discharge during expensive peak hours, and when to hold charge for an expected outage.
Key specs	13.5 kWh usable; up to 11.5 kW continuous output; 89% solar-to-battery-to-home efficiency; 10-year warranty.
Typical price	About **$13,000–$16,500 installed** for one unit in the U.S. in 2026 (hardware about $8,200 plus Gateway about $900, the rest is installation and electrical work).
Installation needs	Tesla Certified Installer only, after a site survey. Usually one day, with a possible short power cut. Floor- or wall-mounted, indoors or outdoors, with code setbacks from windows and doors into living space. Indoor installs need interconnected heat detectors and may need drywall protection, and garage installs need protection from vehicles. The Backup Gateway goes at the main panel, and some homes need a panel upgrade. A building permit and utility interconnection approval are required.

Note: the 30% federal Residential Clean Energy Credit (Section 25D) ended for expenditures after December 31, 2025, so U.S. buyers in 2026 pay full price unless state or utility incentives apply.

Figure 2. Components and how Tesla Powerwall 3 works



3. Sense Home Energy Monitor

A monitor that clamps onto the main electrical feed in the panel and uses machine learning to recognize individual appliances by their electrical signatures, so users can see in real time which devices use the most energy.



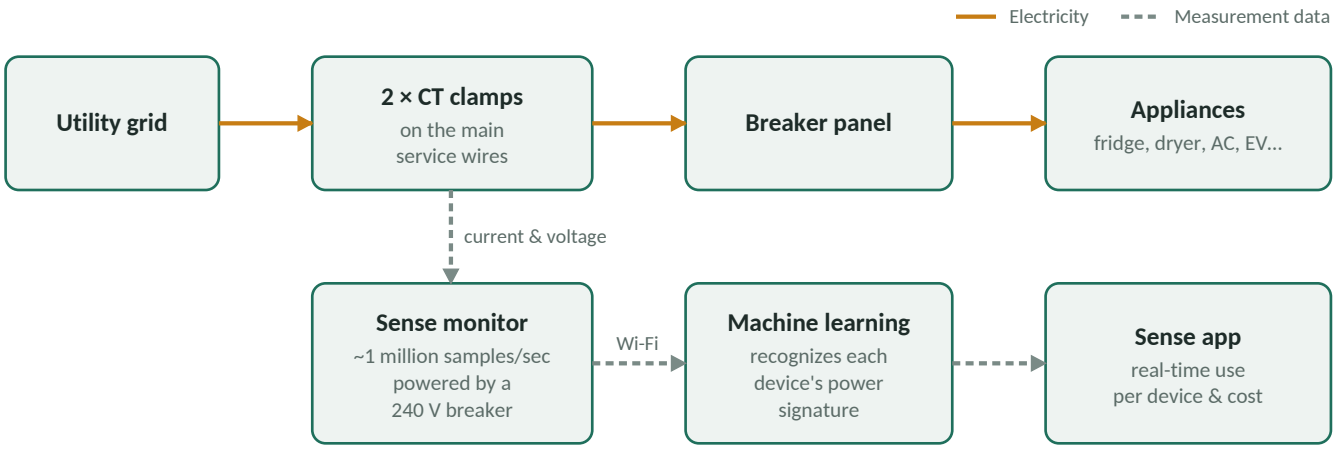
Sense monitor and app.

Manufacturer	Sense Labs, Inc. (Cambridge, Massachusetts)
Components	Monitor unit mounted in or next to the breaker panel; two current-transformer (CT) clamps on the main service wires; power leads connected to a 240 V breaker; Wi-Fi radio. The Sense Solar version adds a second pair of clamps for solar production.

How it works	Measures current and voltage over a million times per second. Machine-learning models pick out each appliance's unique on/off and power pattern, then show device-level usage and cost in the app. Device detection improves over weeks.
Typical price	Was $299 (Sense) and $349 (Sense Solar).
Installation needs	Mounts inside the main breaker panel with the cover off, so a licensed electrician is recommended. Needs a free space for a new 240 V (two-pole) breaker to power the monitor, room for the two clamps around the main wires, and Wi-Fi signal at the panel. If there's no room for the breaker, an electrician has to decide what local code allows. The monitor draws only about 4 W.

Status: Sense stopped selling its consumer monitor on December 31, 2025. Sense technology is now delivered through Sense-enabled utility smart meters, available only where a utility has partnered with Sense. Existing monitors are still supported. Sense itself points buyers to Emporia (item 8) as an alternative.

Figure 3. Components and how Sense Home Energy Monitor works



4. ecobee Smart Thermostat Premium with eco+

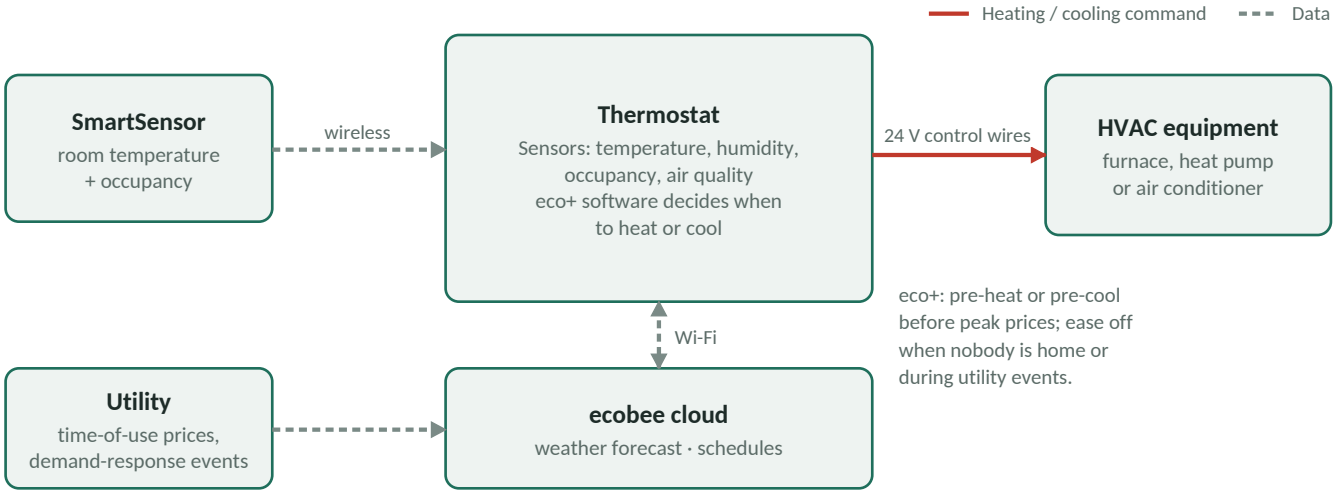
A top choice for optimizing heating and cooling, usually the largest energy use in a home. It adjusts automatically based on occupancy and usage patterns and can take part in utility demand-response programs.



ecobee Smart Thermostat Premium.

Manufacturer	ecobee (a Generac company). Model EB-STATE6-01.
Components	Thermostat with glass face, metal housing and touchscreen; built-in temperature, humidity and occupancy sensors; indoor air-quality monitor; SmartSensor remote room sensor (included); speaker with Alexa or Siri; Power Extender Kit for homes without a C-wire.
How it works	Controls the HVAC system over low-voltage wiring. The eco+ software adds a schedule assistant, smart home/away detection, humidity-adjusted temperature, time-of-use pre-heating or pre-cooling when electricity is cheaper, and community energy savings (demand response).
Key specs	ENERGY STAR certified; ecobee claims up to 26% annual heating and cooling savings compared with a constant 72 °F hold.
Typical price	**$249.99** list; often less through utility marketplaces with instant rebates.
Installation needs	DIY for most homes. Works with most low-voltage (24 V) heating and cooling systems, but not with line-voltage electric baseboard heaters. Needs a C-wire for constant power, or the included Power Extender Kit fitted at the furnace. Place the SmartSensor in the room you use most. Needs home Wi-Fi.

Figure 4. Components and how ecobee Smart Thermostat Premium with eco+ works



5. GE Profile 30-in. Smart Induction Cooktop

Induction cooking is highly efficient and is increasingly chosen over gas. As with all induction cooktops, the pan itself is heated instead of the glass surface. GE adds pan detection, Wi-Fi and guided cooking with an optional smart pan or probe.



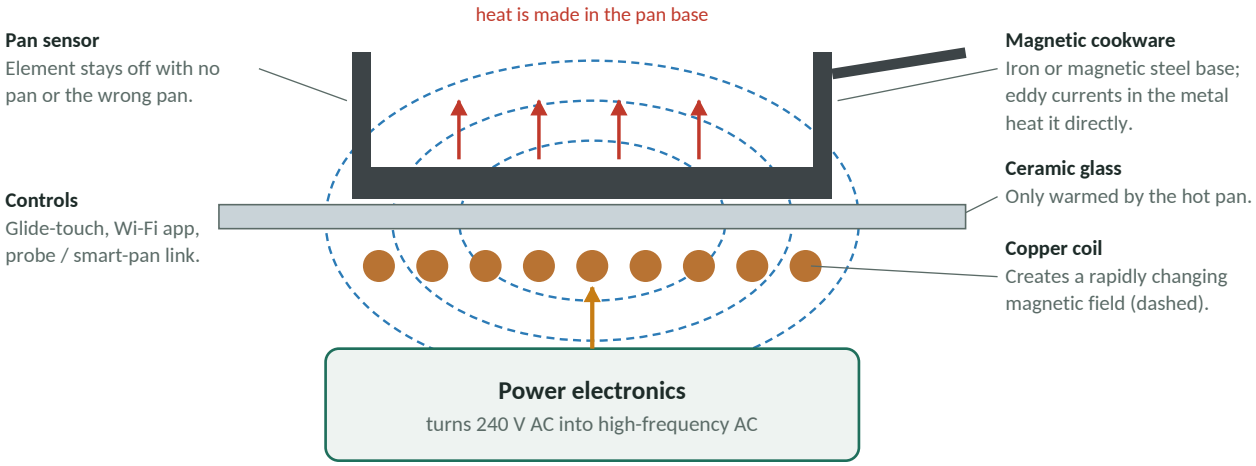


Induction cooktop in use (left); how induction works (right).

Manufacturer	GE Appliances (a Haier company). Model PHP9030DTBB (black) / PHP9030STSS (stainless trim).
Components	Black ceramic glass surface; 4 induction elements (largest 11 in., 3,700 W; smallest 7 in., 2,500 W); glide-touch controls; pan-presence sensor; built-in Wi-Fi (SmartHQ app); Chef Connect; synchronized elements; keep-warm and melt settings; control lock. Optional JXPROBE1 precision cooking probe and Hestan Cue smart pan.

How it works	Copper coils under the glass carry high-frequency alternating current, creating a magnetic field that induces current in magnetic (iron-based) cookware, heating the pan directly. The element turns off automatically when the pan is removed.
Key specs	Fits a 28-1/2 × 19-5/8 in. cutout; requires induction-compatible cookware.
Typical price	MSRP about $2,443–$2,554; retail about **$2,300–$2,340**.
Installation needs	Dedicated 240 V (or 208 V) circuit on a 40 A breaker; the unit is rated 7.4 kW and comes with a 48 in armored cable. Countertop cutout about 28-1/2 × 19-5/8 in (a filler trim kit covers larger openings). Clearances: 3 in below to combustibles (unless over an approved GE wall oven) and 30 in above to unprotected surfaces. Replacing gas usually means an electrician to add the circuit and a plumber to cap the gas line. Needs magnetic cookware.

Figure 5. Components and how GE Profile 30-in. Smart Induction Cooktop works



6. EcoFlow OASIS Energy Management System

Introduced at CES 2025, OASIS is AI software that works with EcoFlow solar and battery systems. It manages power based on household usage, electricity rates, solar production and weather forecasts.

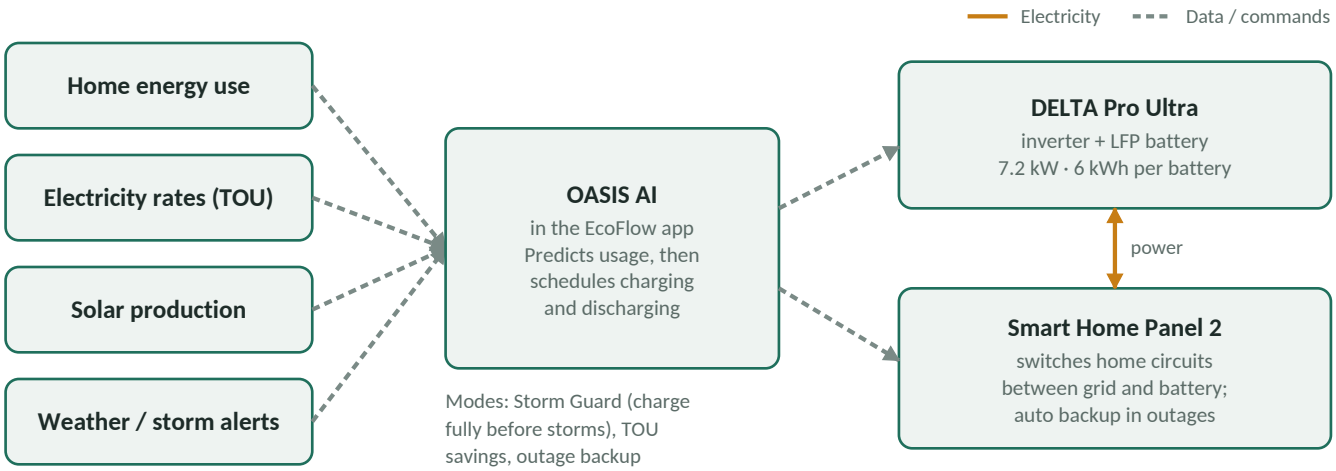




EcoFlow app views and whole-home system.

Manufacturer	EcoFlow (EcoFlow Technology Inc.)
Components	OASIS software in the EcoFlow app, running on EcoFlow hardware. The full whole-home setup in North America is DELTA Pro Ultra (inverter + LFP battery) with Smart Home Panel 2. Also works with smaller DELTA 3 Plus and RIVER 3 Plus units and some third-party devices.
How it works	Analyzes real-time energy use, time-of-use prices, solar generation and weather forecasts, then automates charging and discharging. Modes include Storm Guard (charges fully before severe weather), time-of-use savings and UPS-style backup. Users can also ask questions in plain language.
Key specs	DELTA Pro Ultra: 7.2 kW output and 6 kWh per battery; expandable to 21.6 kW and 90 kWh; 5.6 kW solar input per inverter. Launched for the U.S. market.
Typical price	OASIS is not sold separately. DELTA Pro Ultra (1 inverter + 1 battery) with Smart Home Panel 2 lists around **$4,900–$5,600**, frequently discounted; installation is extra.
Installation needs	DELTA Pro Ultra is plug-in and on wheels, so it needs no installer when used on its own. For automatic whole-home backup, a licensed electrician must install Smart Home Panel 2 (12-circuit, weatherproof enclosure) next to the main panel, with permits as local code requires. OASIS needs the EcoFlow app and Wi-Fi.

Figure 6. Components and how EcoFlow OASIS Energy Management System works



7. Whole House Fans (e.g., QuietCool)

A low-cost way to reduce cooling bills. With windows open, the fan pulls cooler outdoor air through the house and pushes hot air out through the attic. In climates with cool evenings, it can replace air conditioning for much of the year; savings vary widely by climate and use.

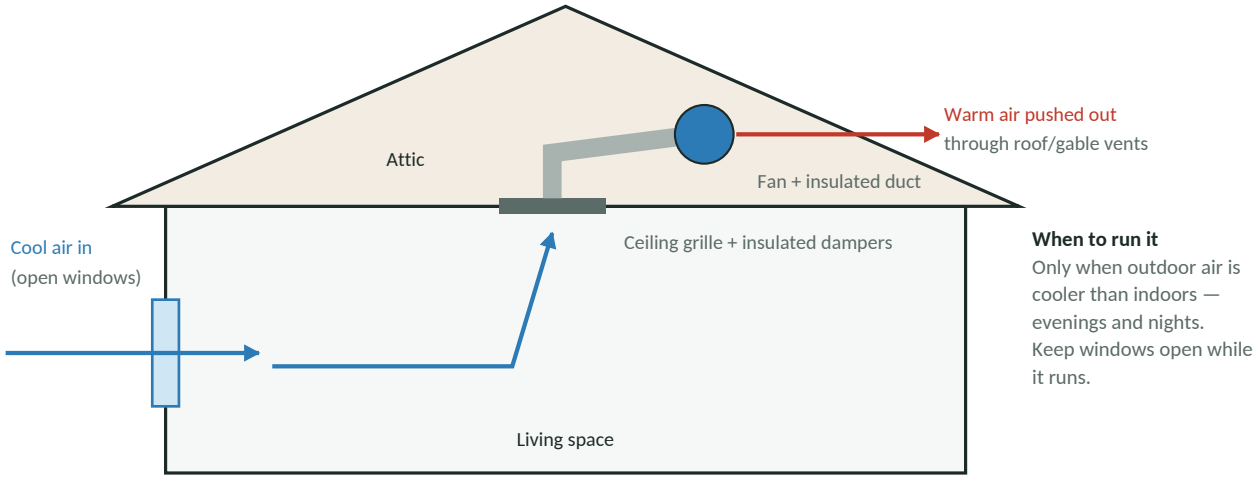


Ducted whole house fans.

Manufacturer	QuietCool (sold through retailers such as Wholehousefan.com, Walmart and Lowe's). Other makers include AirScape and Tamarack.
Components	Fan motor mounted in the attic; flexible insulated duct that dampens noise; ceiling grille with insulated (R5) dampers that close when the fan is off; wall switch or wireless remote; adequate attic venting. Energy Saver models are two-speed.
How it works	Creates a pressure difference that draws outdoor air in through open windows, flushes warm air into the attic and out through roof vents, and cools the home's structure overnight. Effective only when the outdoor air is cooler than the indoor air.
Key specs	Sized at about 2 CFM per sq ft of living space; e.g., ~3,100 CFM covers about 1,550 sq ft. Uses a small fraction of the power of central air conditioning.
Typical price	About **$640–$1,800** for the fan, depending on airflow (1,400–6,900 CFM), plus installation.

Installation needs	Ceiling opening between joists, usually in a central hallway, plus attic space for the fan and duct. The attic needs about 1 sq ft of net free vent area per 750 CFM, so you may need to add vents. Wiring (switch and timer) should be done by an experienced electrician. Safety: always open windows before running the fan so it can't pull exhaust back down gas appliance vents. In humid climates, insulate the damper box.

Figure 7. Components and how Whole House Fans works



8. Emporia Vue 3 Energy Monitor

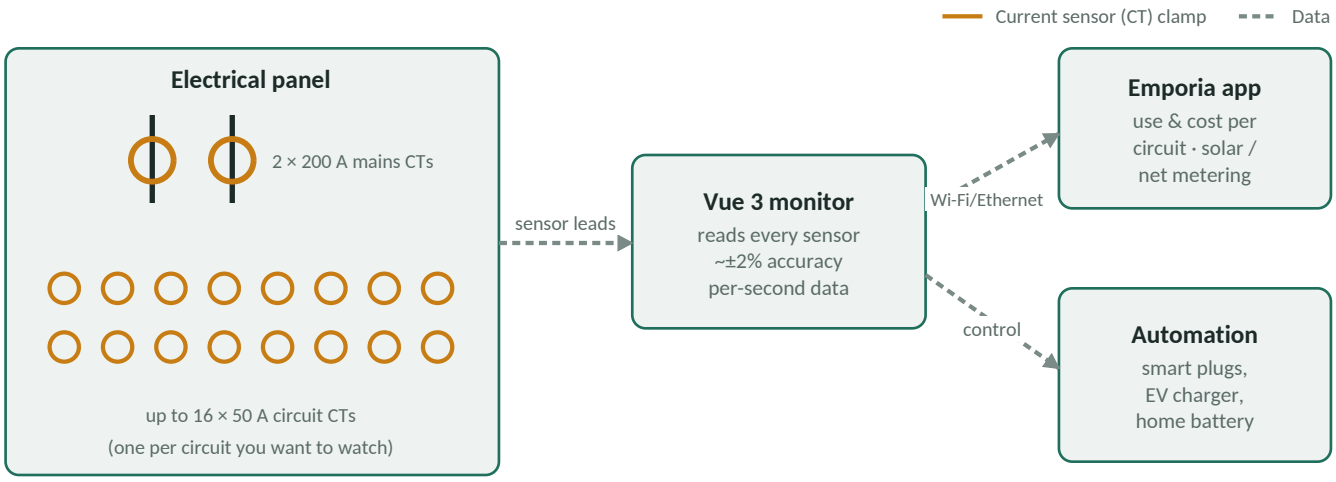
A highly rated, budget-friendly energy monitor with circuit-level tracking, so homeowners can watch high-power loads such as heat pumps and EV chargers individually.



Vue 3 installed in a panel with circuit sensors (left); monitor, sensors and app (right).

Manufacturer	Emporia Corp. (Littleton, Colorado). Launched March 2024.
Components	Vue 3 monitor unit; Wi-Fi antenna and cable (wired Ethernet also supported); two 200 A main current sensors; up to sixteen 50 A circuit sensors; circuit labels.
How it works	Clamp-on current sensors measure the mains and individual branch circuits. The monitor samples them thousands of times per second and sends per-second data to the Emporia app, which shows usage and cost per circuit, tracks solar/net metering, and can automate Emporia smart plugs, EV chargers and batteries.
Key specs	About ±2% accuracy; single-phase 120/240 V; separate 3-phase version available; UL certified.
Typical price	**$99** (whole home only), **$149** (8 circuit sensors), **$199.99** (16 circuit sensors).
Installation needs	DIY if you're comfortable working inside a breaker panel; otherwise hire an electrician, or if your code, HOA or insurer requires one. The wire harness connects to a new single- or two-pole breaker, or to an existing breaker using pigtails. Needs room in the panel for the sensors, plus Wi-Fi or Ethernet. Panels with only busbar access need Emporia's flexible sensors.

Figure 8. Components and how Emporia Vue 3 Energy Monitor works



9. Midea Cube 50-Pint Dehumidifier (R-32 refrigerant)

An efficient home dehumidifier that uses R-32 refrigerant, which has a much lower global-warming potential than the older R-410A. Its expandable tank holds up to 3 times more water than a standard bucket.

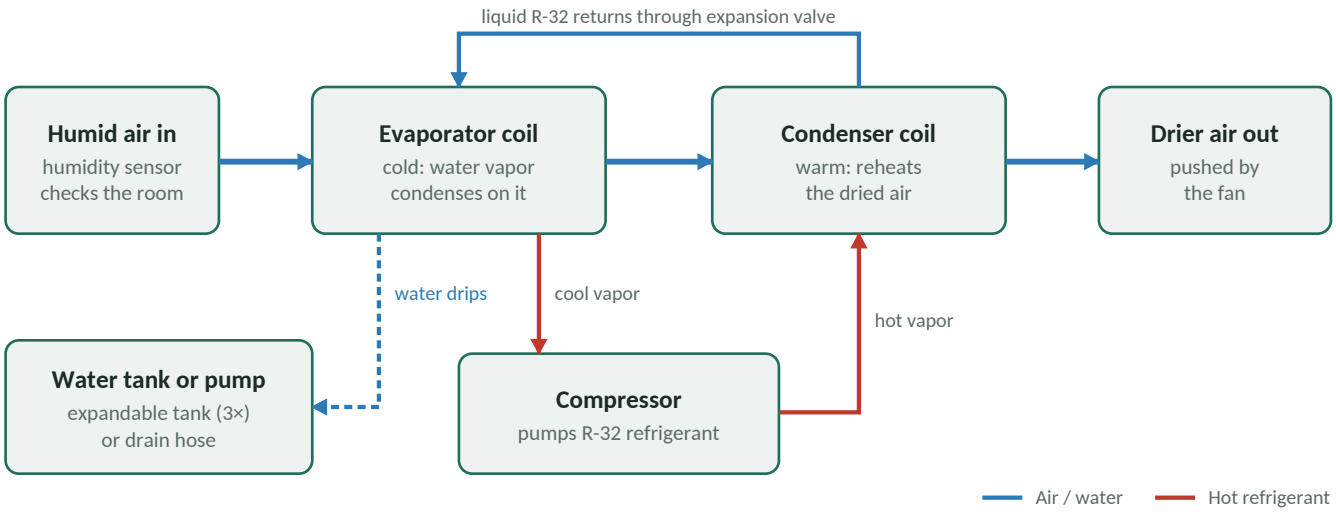


Midea Cube dehumidifier with carry handle and top air outlet.

Manufacturer	Midea Group. Models MAD50S1QWT (standard) and MAD50PS1QWT (with built-in pump).
Components	Compressor-driven refrigeration circuit (R-32) with evaporator and condenser coils; fan; humidity sensor; expandable “lift-and-twist” water tank; optional built-in condensate pump; drain hose; Wi-Fi control (Midea Air app, Alexa and Google Assistant); casters.
How it works	The fan pulls humid air across the cold evaporator coil, where moisture condenses and drips into the tank or drain. The air then passes over the warm condenser coil and returns to the room drier. The unit cycles on and off to hold the humidity level you set.

Key specs	Removes up to 50 pints/day; rated for spaces up to 4,500 sq ft (manufacturer claim); ENERGY STAR Most Efficient 2023.
Typical price	About **$260–$270** (standard); pump version lists around $300–$345 and is often on sale for about $190–$230.
Installation needs	No installation: plugs into a standard grounded 115 V outlet. Leave clearance around it (Midea manual: about 8 in at front and sides, 16 in at top and rear). For continuous draining, place it near a floor drain or sink, or use the pump model to send water up to a sink or out a window. Close the room's doors and windows for best results.

Figure 9. Components and how Midea Cube 50-Pint Dehumidifier works



10. Solar-Powered Outdoor Security Lighting

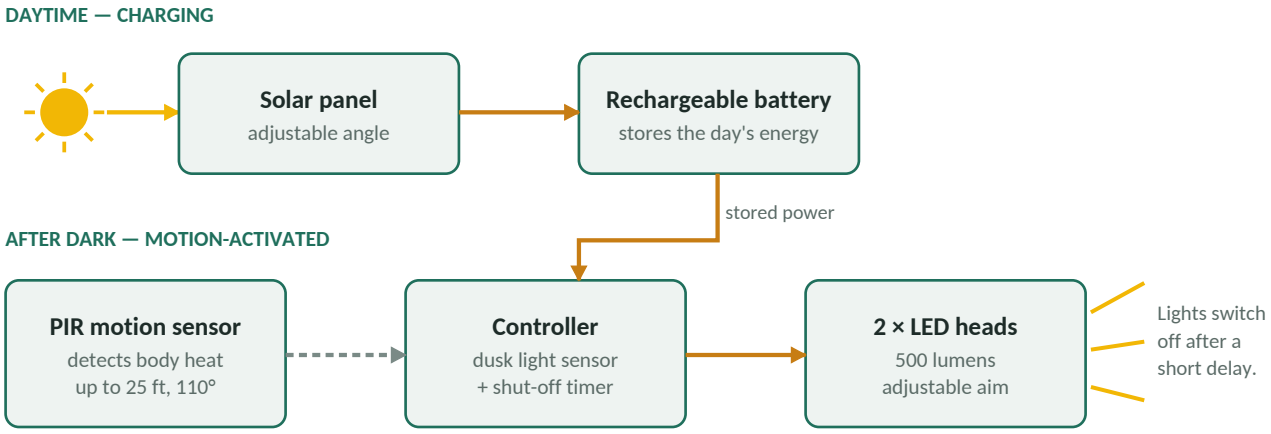
Motion-activated lights (and some cameras) that run on solar power, adding home security with no wiring and no electricity cost. Example: Bell+Howell Bionic Spotlight Duo.



Solar motion floodlight (left); Bell+Howell Bionic Spotlight Duo (right).

Manufacturer	Bell+Howell (Bionic Spotlight Duo, model 7782). Many other brands sell similar lights.
Components	Adjustable solar panel; built-in rechargeable battery; two adjustable LED heads; passive infrared (PIR) motion sensor; light sensor for dusk-to-dawn operation; ground stake or wall mount; weather-resistant housing.
How it works	The panel charges the battery during the day. After dark, the PIR sensor detects body heat moving within range and turns the LEDs on at full brightness, then switches them off after a short delay.
Key specs	500 lumens; 6 W LEDs; 25 ft motion detection range; 110° coverage.
Typical price	About **$30** per light.
Installation needs	DIY with no wiring. Mount on a wall or stake where the solar panel gets several hours of direct sun a day, and aim the motion sensor at the path or door you want covered. Wipe the panel now and then so it charges fully.

Figure 10. Components and how Solar-Powered Outdoor Security Lighting works



11. Rooftop Solar with Enphase IQ8 Microinverters

The core of most natural-energy homes. Solar panels turn sunlight into electricity, and Enphase IQ8 microinverters convert it to household AC right at each panel. Production is tracked panel by panel, and shade on one panel doesn't drag down the rest.



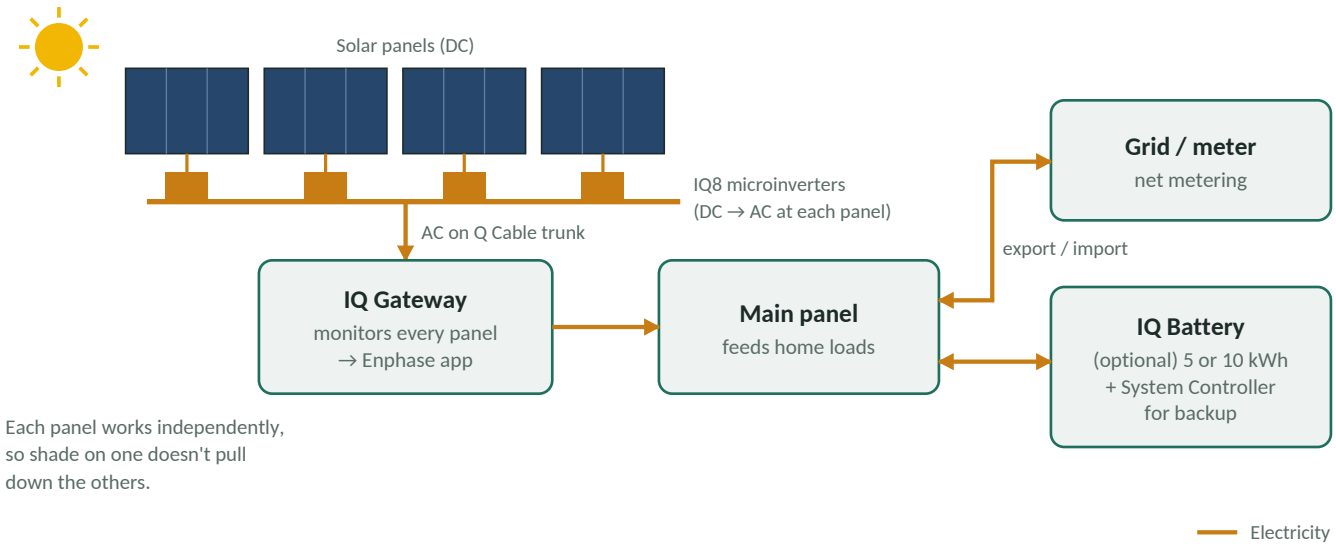
Solar panel kit with Enphase microinverters, combiner box, mounting hardware and monitoring app.

Manufacturer	Enphase Energy (microinverters, gateway, batteries). The panels come from separate makers such as Qcells or REC, chosen by the installer.
Components	Solar panels on roof racking; one IQ8 microinverter under each panel (IQ8+, IQ8M, IQ8A or IQ8HC, matched to panel wattage); Q Cable trunk; IQ Gateway for monitoring; optional IQ System Controller and IQ Battery (5 or 10 kWh, LFP) for backup; Enphase app.

How it works	Each microinverter runs its panel at its best operating point and converts DC to AC on the roof, so no high-voltage DC runs across the house. AC feeds the home first. Surplus goes to the grid (net metering, where offered) or to a battery. With the System Controller, IQ8 can keep running on sunlight during an outage, even without a battery (Sunlight Backup).
Key specs	IQ8 output 300–384 W per unit; about 96.5–97% CEC efficiency; 25-year warranty. A typical U.S. home system is about 12 kW.
Typical price	About **$2.60 per watt** installed, or roughly **$31,000** for a 12 kW system before incentives (EnergySage 2026 average). Microinverters alone are about $170–$200 per panel.
Installation needs	Licensed or NABCEP-certified solar installer. Needs a structurally sound roof with plenty of life left, ideally facing south or west with little shade. Requires a building permit and utility interconnection or net-metering approval. The IQ Gateway goes within about 30 ft of the main panel, and some homes need a main panel upgrade. Typically 1–3 days on the roof, plus weeks of permitting.

Note: the 30% federal tax credit for homeowner-owned solar (25D) ended after December 31, 2025. Leases and power-purchase agreements (PPAs) can still pass on a commercial credit, and state or utility incentives vary.

Figure 11. Components and how Rooftop Solar with Enphase IQ8 Microinverters works



12. Mitsubishi Hyper-Heating Ductless Mini-Split Heat Pump

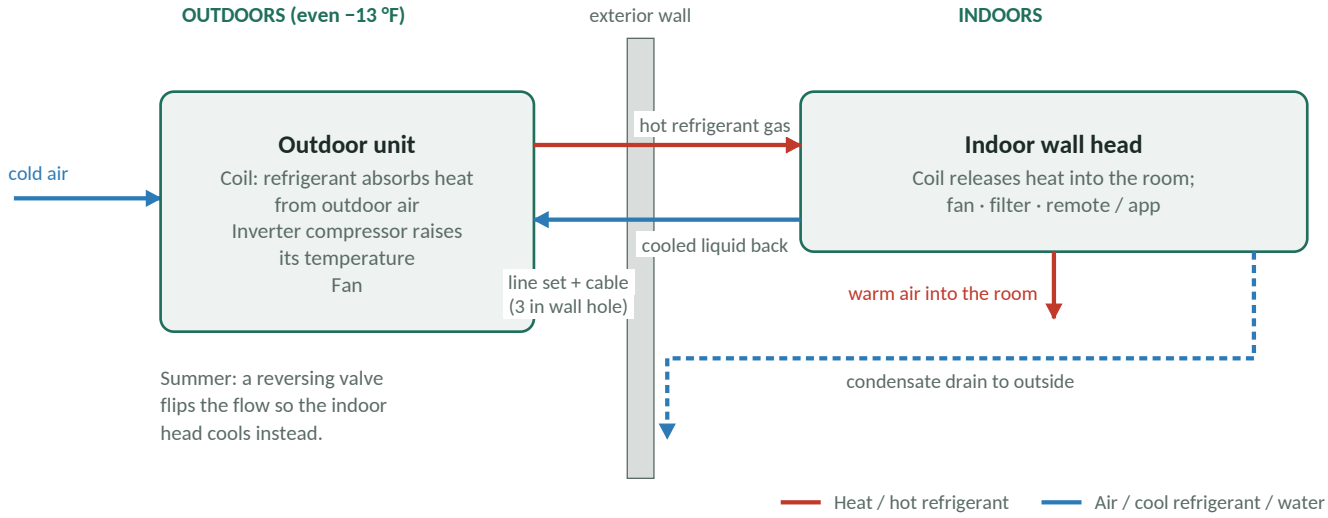
A cold-climate heat pump that pulls heat out of outdoor air to warm the home in winter, then reverses to cool it in summer. It replaces or supplements oil, propane or electric-resistance heat, and needs no ductwork.



Mitsubishi Electric outdoor unit and indoor wall head, shown with a line-set installation kit.

Manufacturer	Mitsubishi Electric (Hyper-Heating INVERTER, “H2i”, line).
Components	Outdoor unit with variable-speed inverter compressor (flash-injection), fan and coil; one or more indoor wall heads (or ducted or ceiling units) with fan, coil and filter; refrigerant line set plus power/communication cable through a 3 in wall hole; condensate drain; remote, wall controller or app.
How it works	Refrigerant absorbs heat from outdoor air at the outdoor coil, even well below freezing. The compressor raises its temperature and the indoor head releases the heat into the room. In summer a reversing valve sends heat the other way. The inverter compressor speeds up and slows down instead of switching on and off.
Key specs	Heats down to −13 °F. Example 1.5-ton system: SEER2 19, HSPF2 12.5, about 70% of rated heating capacity at 5 °F, COP 1.8 at −5 °F.
Typical price	Mitsubishi single-zone systems run about **$1,200–$5,000 installed** (average about $2,800 for 9,000–12,000 BTU). Hyper-Heating models sit at the higher end, and multi-zone or whole-home systems cost considerably more.
Installation needs	Licensed HVAC contractor, ideally Mitsubishi Diamond-certified, with a room-by-room load calculation first. Needs a wall or pad spot outdoors, raised above snow line in cold areas; a line set (often up to 15 ft included); a condensate drain or pump; and a dedicated 240 V circuit. Check whether the panel has room. Single-zone installs take about 3–8 hours; multi-zone takes 1–2 days. Permits $0–$400.

Figure 12. Components and how Mitsubishi Hyper-Heating Ductless Mini-Split Heat Pump works



13. Geothermal Heat Pump (e.g., WaterFurnace 7 Series)

Uses the steady temperature underground as its heat source in winter and its heat sink in summer. This makes it one of the most efficient ways to heat and cool a home, and it can also preheat hot water.



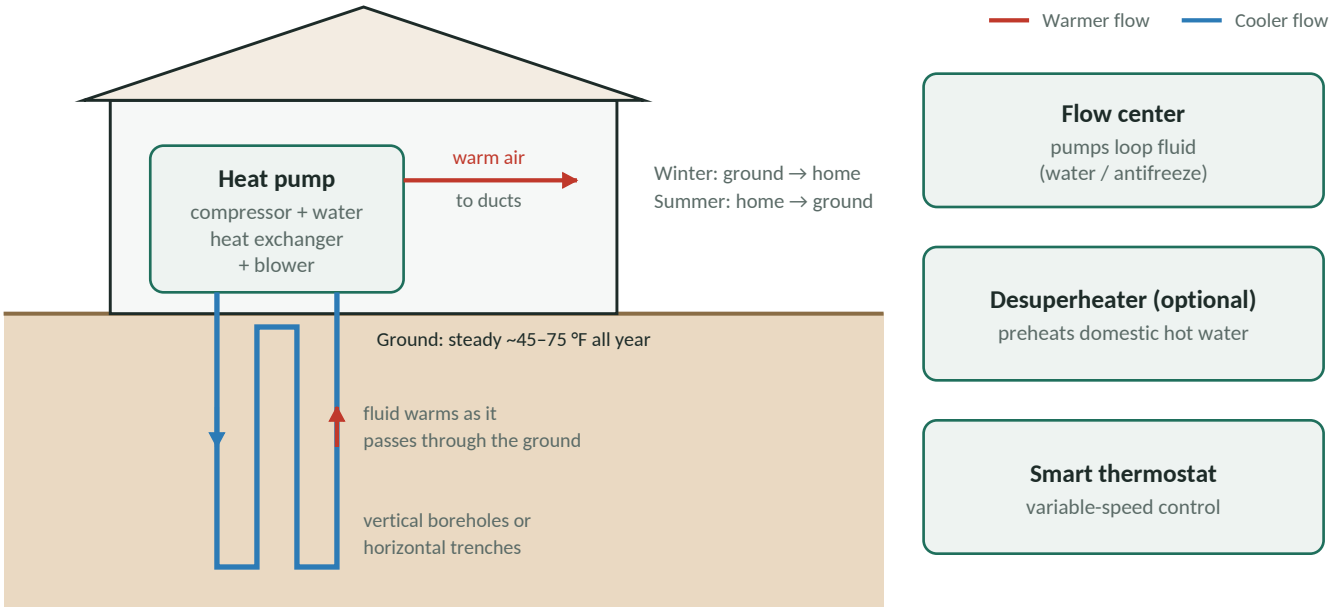
WaterFurnace 7 Series with ground-loop connections (left); cabinet opened to show blower, controls and refrigerant circuit (right).

Manufacturer	WaterFurnace International (7 Series variable-speed). Others include ClimateMaster, Bosch and Dandelion Energy.
Components	Indoor heat pump cabinet (variable-speed compressor, refrigerant-to-water heat exchanger, variable-speed blower); ground loop of buried HDPE pipe in vertical boreholes or horizontal trenches, filled with water/antifreeze; flow-center pumps; smart thermostat and controls; optional desuperheater for hot water; existing or new ductwork.

How it works	A few feet down, the ground stays roughly 45–75 °F all year. Fluid in the loop picks up that heat in winter, and the heat pump concentrates it and blows warm air through the ducts. In summer the process reverses and indoor heat goes into the ground. Because the source temperature is mild, it runs far more efficiently than an air-source unit on very hot or cold days.
Key specs	Typical home size 3–5 tons; ENERGY STAR certified; variable speed for long, quiet, low-power running.
Typical price	About **$14,000–$34,000** total: indoor unit $7,000–$13,000, installation labor $3,000–$9,000, ground loop $4,500–$20,000 depending on lot size and geology.
Installation needs	Specialist geothermal installer plus a driller (vertical) or excavator (horizontal). Horizontal loops need a large yard; vertical boreholes suit small lots but cost more. Requires drill-rig access, soil or geology checks, and permits (some areas need well-drilling permits). Indoor space is about the size of a furnace. Takes about 2–5 days, and duct changes may be needed.

Note: geothermal heat pumps were covered by the 30% 25D federal credit, which ended after December 31, 2025. Check state and utility programs.

Figure 13. Components and how Geothermal Heat Pump works



14. Heat Pump Washer-Dryer: GE Profile UltraFast Combo

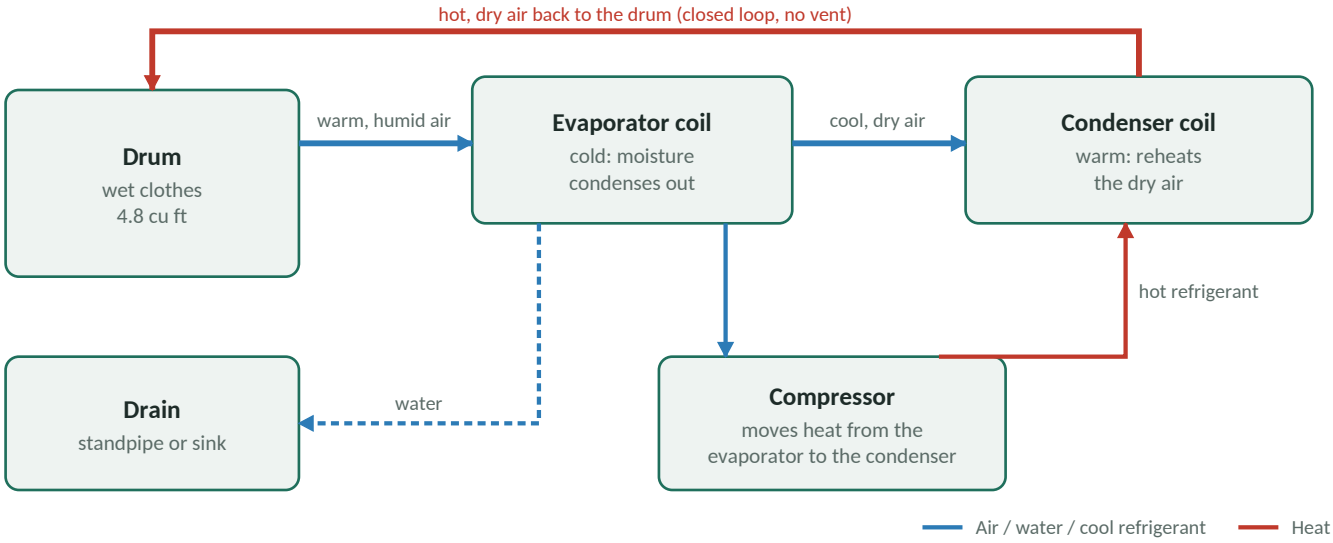
An all-in-one washer and dryer that dries with a heat pump instead of a resistance heater. It uses far less electricity, needs no dryer vent, and runs on a standard 120 V outlet.



GE Profile UltraFast Combo washer-dryer.

Manufacturer	GE Appliances (a Haier company). Model PFQ97HSPV (several colors).
Components	4.8 cu ft stainless drum; washer motor; sealed heat pump drying system (compressor, evaporator and condenser coils, fan); condensate drain path; Wi-Fi (SmartHQ app); fill and drain hoses.
How it works	Air circulates in a closed loop. It picks up moisture from the clothes, passes over the cold evaporator coil where the water condenses and drains away, then is reheated on the warm condenser coil and sent back into the drum. Nothing is vented outside, and the heat is recycled rather than thrown away.
Key specs	ENERGY STAR; about 136 kWh/year; GE claims 50% more efficient drying than a standard ENERGY STAR electric dryer; a full load washes and dries in about 2 hours; 28 in W × 32 in D × 46.7 in H; 302 lb.
Typical price	**$2,999** list; seen on sale around **$2,100**.
Installation needs	Standard grounded 120 V outlet on a 15 A circuit, with no 240 V circuit or dryer vent needed. Needs hot and cold water hookups and a drain standpipe or sink. A 28 in wide spot fits closets and small laundry areas. The floor must be level and solid enough for about 300 lb plus water. Leave room to open the door (54.5 in depth with door open).

Figure 14. Components and how Heat Pump Washer-Dryer: GE Profile UltraFast Combo works



15. Solar-Aware EV Charger: Emporia Level 2

A home EV charger that can be set to charge only from excess rooftop solar, or to wait for off-peak rates. The car's battery becomes a way to use your own solar power instead of exporting it.

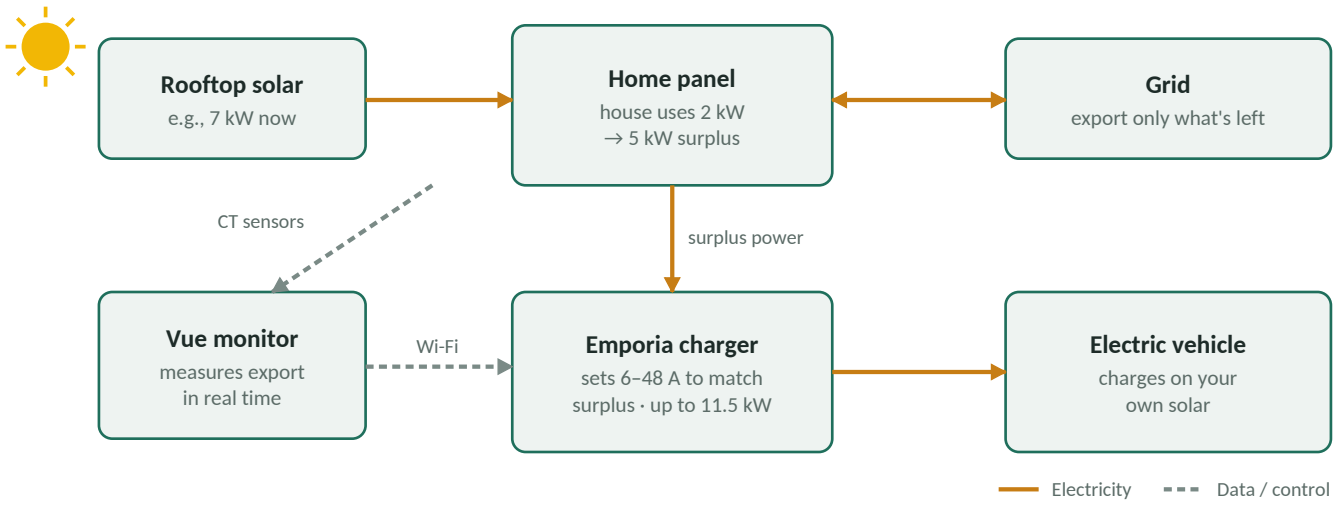


Emporia Level 2 EV charger with the Emporia app's charging schedule.

Manufacturer	Emporia Corp. (Littleton, Colorado), the same company as the Vue monitor in item 8.
Components	Level 2 charger in a NEMA 4 (weatherproof) enclosure; 24 ft cable with J1772 or NACS (Tesla) connector; hardwired or NEMA 14-50 plug version; Wi-Fi; Emporia app. Bundles add a Vue energy monitor and sensors.

How it works	The Vue monitor sees how much solar power is being exported. The charger raises or lowers its current (6–48 A) to soak up just that surplus, so the car runs on your own solar. It can also follow time-of-use schedules and cut back when the house is near its electrical limit.
Key specs	Up to 11.5 kW hardwired (48 A) or 9.6 kW plug-in (40 A); adjustable current; 3-year warranty.
Typical price	About **$599** for the charger; installation typically **$800–$3,000** depending on the home's wiring.
Installation needs	Licensed electrician recommended. Needs a dedicated two-pole circuit: 60 A for 48 A charging, or 50 A for 40 A charging and the plug-in version. Panel capacity matters, though load management can help avoid an upgrade. Indoor or outdoor mounting near the parking spot. 2.4 GHz Wi-Fi. Solar-only charging also needs the Vue monitor installed in the panel.

Figure 15. Components and how Solar-Aware EV Charger: Emporia Level 2 works



16. Solar Generator: Anker SOLIX F3800 Plus

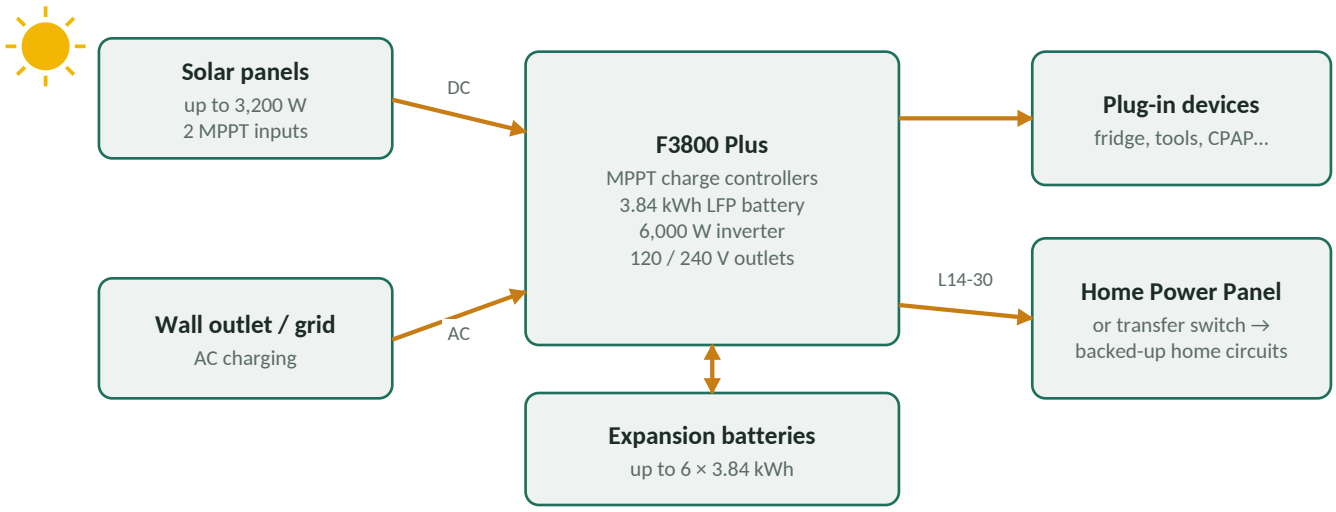
A large portable battery and inverter that charges from solar panels or the wall and can back up key home circuits. It's an entry point to solar storage without a permanent roof system.



Anker SOLIX F3800 Plus solar generator kit with solar panels.

Manufacturer	Anker Innovations (SOLIX brand).
Components	Power station with 3.84 kWh LFP battery, 6,000 W inverter, two solar MPPT inputs and 120 V / 240 V outlets (including NEMA L14-30); portable or rooftop solar panels; optional BP3800 expansion batteries; optional Home Power Panel or transfer switch for home backup; Anker app.
How it works	Solar panels charge the LFP battery through built-in MPPT controllers, and it can also charge from a wall outlet. The inverter turns stored DC into 120/240 V AC for devices plugged in directly, or for selected home circuits through a transfer switch or Home Power Panel during outages.
Key specs	3.84 kWh; 6,000 W output (120/240 V); up to 3,200 W solar input; expandable with up to 6 expansion batteries (to about 53.8 kWh); 3,000+ cycles; 136.7 lb.
Typical price	**$2,499** list; seen on sale around **$2,090** (mid-2026). Solar panels, expansion batteries and Home Power Panel cost extra.
Installation needs	None for plug-in use or adding expansion batteries; Anker calls these plug-and-play. Connecting to home circuits needs a transfer switch, inlet box or Anker Home Power Panel installed by a licensed electrician. Solar panels can sit on the ground or be mounted. Keep it in a dry, ventilated spot; it's heavy but on wheels.

Figure 16. Components and how Solar Generator: Anker SOLIX F3800 Plus works



Key 2025 Trends

- **AI integration:** Many products now use machine learning to manage energy automatically, such as scheduling a dishwasher or dryer for times when solar power is available.
- **Heat pump technology:** Heat pumps are one of the fastest-growing technologies for heating, cooling and water heating, and ENERGY STAR promotes them as among the most efficient all-electric options.
- **Solar self-consumption:** Systems that let homeowners store and use their own solar power (such as Tesla Powerwall or EcoFlow whole-home systems) are being adopted more widely.